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Real-time prediction of high-speed rail pantograph-catenary contact force using deep learning framework

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ABSTRACT

In high-speed railway systems, the contact force is a crucial metric for describing the interaction performance between the pantograph and the catenary. However, measuring the contact force directly is challenging compared to other measurable quantities such as panhead acceleration and uplift. Leveraging advancements in artificial intelligence, this work proposes a deep learning framework to predict the contact force time history based on panhead acceleration and uplift data. Canonical recurrent neural network models, including vanilla RNN, LSTM and GRU are employed to model and predict the contact force. To enhance efficiency by reducing model parameters and extracting sequential feature maps, we propose a hybrid GRU model integrated with a one-dimensional convolutional neural network, called CNN-GRU. The database for this study is built using a validated and reliable finite element model of the pantograph-catenary system, incorporating variations in structural parameters, speed levels and cut-off frequencies. Four training strategies are designed to evaluate prediction performance across these variations. The results demonstrate the potential of using deep learning to predict contact force based on panhead acceleration/uplift, with some limitations in predicting across different cut-off frequencies. While vanilla RNN, LSTM and GRU models achieve acceptable prediction accuracy, the proposed CNN-GRU outperforms them, particularly in terms of prediction consistency across varying train speeds.

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1. Introduction

The pantograph-catenary system serves as the backbone for powering trains in electric railways, as illustrated in Figure 1. This system facilitates the transfer of electrical current from the catenary to the train, enabling its high-speed service [1]. The pantograph, mounted atop the train, maintains contact with the catenary wire, ensuring a continuous supply of power [2,3]. The interaction dynamics between the pantograph head and the contact wire are crucial for the efficient operation of the high-speed railway system. One of the key metrics used to evaluate this interaction is the contact force between the pantograph head and

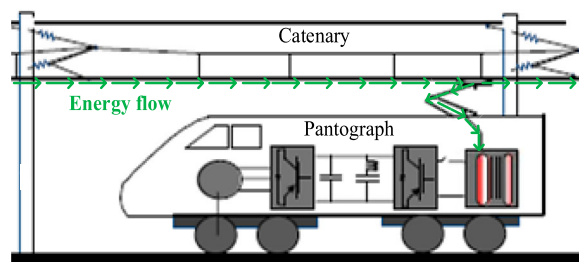


Figure 1. Pantograph-catenary system used for powering trains in electric railways.

the contact wire [4]. This force directly influences the quality of current collection, which is essential for ensuring a stable and reliable power supply to the train.

1.1. Problem statement

The paramount metric for assessing the interaction dynamics of the pantograph-catenary system is the contact force between the pantograph head and the contact wire. This force directly reflects the quality of current collection, a critical aspect outlined in assessment standards such as EN 50367 [5]. Stability and minimal fluctuations in contact force are pivotal criteria, hence making the standard deviation of contact force the primary measure for evaluating current collection quality. Consequently, accurate measurement of contact force holds significant importance in the operational efficiency of high-speed rail systems [6].

In compliance with EN 50317 [7], protocols are established for gathering the contact forces between the pantograph and catenary. Typically, specialised inspection trains, equipped with instrumented pantographs, traverse high-speed networks to gather this crucial quantity. As illustrated in Figure 2, pressure sensors are strategically affixed to each of the four support points of the panhead to capture the force exerted between the panhead and the framework. Additionally, two vertical accelerometers are positioned on each pantograph strip to record the inertial forces generated by the strips. The sensor placement configurations are detailed in Figure 2, denoting mounting points 1–4 for the spring force sensors and 5–6 for the accelerometers. The measured contact force can be mathematically

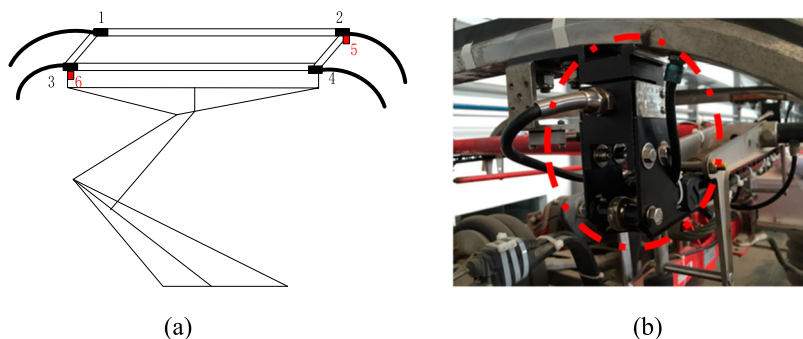


Figure 2. Installation of sensors on the pantograph head: (a) configurations of installing sensors; (b) Photo of pantograph head with sensors.

expressed as follows:

$$f_c^{\text{meas}} = (f_{\text{sp}}^1 + f_{\text{sp}}^2 + f_{\text{sp}}^3 + f_{\text{sp}}^4) + m_{\text{strip}}(a_{\text{strip}}^1 + a_{\text{strip}}^2) + f_{\text{aero}} \quad (1)$$

in which, the contact force f_c^{meas} is calculated as the sum of spring forces, inertial forces and aerodynamic contribution f_{aero} . Four spring forces f_{sp}^{1-4} are obtained from the vertical force sensors mounted between the panhead and its support, while the inertial forces are computed using two measured vertical accelerations a_{strip}^{1-2} of the pantograph strips and their known masses m_{strip} . The measurement framework highlights a noticeable dependence of the measured contact force on the panhead acceleration and uplift, which are much more convenient to be obtained. While many railway operators worldwide have embraced acceleration or uplift measurement due to its practical implementation, gauging contact force poses a challenge, given the impracticality of placing pressure sensors directly on the contact surface [8,9]. Consequently, the approach described above offers an indirect means to approximate the actual contact force, albeit with limited accuracy. Thus, there's a compelling need to develop a predictive model for contact force using measured panhead acceleration and uplift. This research frontier has garnered considerable attention, as evidenced by [10]. Within this domain, various models have emerged to forecast contact force or its statistics based on acceleration data. This paper follows this novel endeavour and integrates deep learning methodologies into constructing a highly efficient prediction model for contact force using the panhead acceleration and uplift.

1.2. Literature review

The investigation of pantograph-catenary dynamics has garnered significant attention from scholars in both industry and academia, with the rapid development of the high-speed rail network. Various mathematical modelling methods have been developed to accurately reproduce realistic pantograph-catenary interaction behaviours. According to the world benchmark established in 2015 [11], the most commonly used method for representing the catenary is the finite element (FE) approach [12]. Recently, several FE models of catenary have been developed based on the Euler-Bernoulli beam theory and commercial software [13]. As the pantograph-catenary system works in a complex environment, the modelling of real-life disturbances is of great importance to evaluate the service performance and ensure operational safety [14]. The crosswind along the catenary has been modelled using the empirical spectrum and incorporated into the numerical model [15–17]. The common disturbances from the carbody vibration induced by the track irregularities have been included in the assessment of the interaction performance [18,19]. To create a more realistic simulation environment, hardware-in-the-loop test rigs have been developed to couple a mimic catenary model with a realistic pantograph, which has the potential to test active pantographs equipped with controllers and actuators for regulating contact force [20–22].

The catenary serves as both the electric conduit for the current and the mechanical slide for the pantograph. The premise of maintaining a good current collection quality is the stable contact of the pantograph-catenary [23–25]. Thus, the most important index to evaluate the current collection quality is the contact force between the contact wire and the panhead. As described above, the direct measurement of the contact force is almost impossible. In contrast, the acceleration and uplift are more likely to be obtained, which contains

the fluctuation component of the interaction behaviour. Normally, the panhead acceleration has a higher-frequency resolution, and can be used to detect some short-wavelength irregularities, such as the local impact as reported in [9]. In [26], an EEMD (Empirical Ensemble Mode Decomposition) and HHT (Hilbert Huang Transform)-based approach is proposed to identify and localise the contact wire irregularity with short wavelength. To accurately capture the contact force, a measurement scheme is proposed using measured strains in the panhead by Fibre Bragg grating sensors [27]. The use of the contact force to do damage detection is also an important development trend in the current research. In [28], the contact force against the static height of Shinkansen contact wires and their application in overhead line diagnosis are reported. The application of the contact force for predicting the failure of catenary components was investigated via a numerical approach in [29]. In [30], a 1D-CNN network is proposed to detect and localise the defective droppers in railway catenary using the interaction responses. Apart from the contact force, the vision-based diagnosis method is also the state-of-the-art trend for ensuring the safety of the pantograph-catenary operation. Using the mobile scanning data, a detection, localisation and classification approach of the catenary system is proposed in [31]. In [32], a destruction and reconstruction learning method is proposed to diagnose the defect of the split pins of the catenary.

The challenge of predicting contact force in the pantograph-catenary (PC) system has led to two main research directions. The first focuses on indirect measurement techniques and surrogate modelling to estimate contact force using more accessible signals, such as panhead acceleration and uplift. Gregori et al. [10] proposed using artificial neural network models to predict the contact force statistics using the panhead acceleration. Time-series data like contact force, comprising sequential observations over time, demands specialised analytical methods to effectively capture temporal dependencies, trends and seasonal patterns inherent in such data. The second direction encompasses methods that handle sequential time-series modelling in the broader context of railway systems. Traditional statistical approaches like autoregressive integrated moving average (ARIMA) models [33] and Exponential Smoothing techniques, including Holt-Winters methods, have been foundational for forecasting by modelling linear relationships and accommodating trends and seasonality. State Space Models, such as Kalman Filters and Hidden Markov Models [34], offer probabilistic frameworks for modelling dynamic systems, especially when dealing with unobservable states. With the evolution of machine learning, methods like support vector regression [35] and ensemble algorithms – such as Random Forests, Gradient Boosting Machines and their optimised versions like XGBoost and LightGBM – have been applied to time-series data to handle non-linear relationships and complex patterns for tasks like forecasting and anomaly detection. Clustering techniques utilising dynamic time warping and K-nearest neighbours [36] have also been employed for classifying and measuring similarities in time-series data. The advent of deep learning [37] has introduced powerful models like recurrent neural networks (RNNs) [38], Particularly long short-term memory (LSTM) [39] networks and gated recurrent units (GRUs) [40], which excel at capturing long-term dependencies in sequential data. Convolutional neural networks (CNNs) [41] have been adapted to extract local temporal features, and transformer models with attention mechanisms have revolutionised time-series forecasting by effectively modelling long-range dependencies without reliance on recurrence. Hybrid models that combine traditional statistical methods with neural networks aim to leverage the

strengths of both approaches, enhancing predictive performance. Advanced techniques like temporal fusion transformers [42] and temporal convolutional networks [43] further improve forecasting capabilities for multivariate time-series data.

1.3. Contribution of this work

The above literature review highlights that predicting the contact force in high-speed rail pantograph-catenary systems has become a focal point, primarily due to the challenges in measuring accurate contact force directly. Panhead acceleration and uplift, which are much easier to obtain, offers a promising alternative for such assessments. However, most previous studies have only established statistical relationships between acceleration and contact force, without predicting the time history of the contact force – a much more crucial element containing useful information. To address this gap, this paper explores the potential of using panhead acceleration and uplift to predict the contact force through deep learning neural networks. A deep learning framework with common time history prediction methods, like LSTM, RNN and GRU is proposed to capture the non-linear relationship between panhead acceleration/uplift and contact force time histories. A CNN-GRU combined prediction approach is also proposed to address the characteristics of the current problem. A reliable numerical model of the pantograph-catenary system is used to construct the database with different structural parameters, train speeds and cut-off frequencies, enabling the test of the real-time contact force prediction from panhead acceleration/uplift. Several classic neural networks are compared in terms of their prediction accuracy with different training strategies.

2. Deep learning framework

The applications of GRU, LSTM and RNN to time-series prediction in different engineering scenarios have been extensively reported [44–48]. In this work, we propose a deep learning framework designed to address the challenges of modelling contact force (sequential data) by effectively utilising both displacement and acceleration inputs. The approach integrates a one-dimensional convolutional neural network (1D-CNN) to extract features from the input data while simultaneously reducing computational complexity. The extracted features are then processed by a recurrent neural network (RNN) based on gated recurrent units (GRUs), which are employed to capture the hidden temporal dependencies within the sequential data. Finally, a fully connected neural network (FNN) layer is used to predict the contact force over time, leveraging the learned representations from the GRU. This architecture efficiently combines feature extraction, sequential learning and prediction in a unified model, as illustrated in Figure 3.

In Figure 3, the notation $a^{\rightarrow<0>}, \dots, a^{\rightarrow<Tx+1>}$ denotes the sequence of activation vectors generated by the unrolled RNN/GRU/LSTM during forward propagation, where the superscript

$<t>$ indicates the activation at time step t . The input term X_{t-m} represents the feature at time index $t-m$, with m denoting the window size (i.e. the number of past samples included in the input sequence). Thus, the model receives a sequence of m consecutive samples, from X_{t-m} to X_{t-1} , as its input at time t . We present the details of the model as the following:

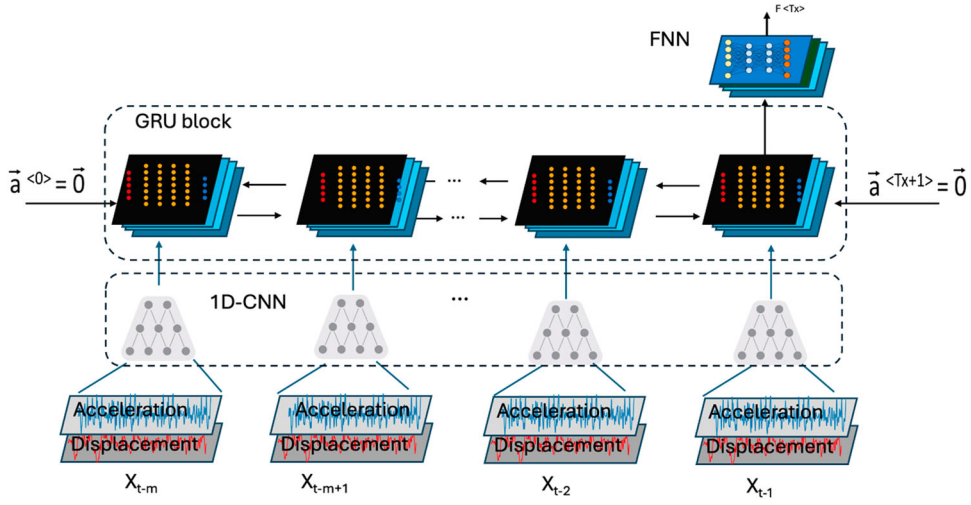


Figure 3. RNN for pantograph-catenary system.

The **Input Block** is responsible for sampling the input data, which includes two channels: acceleration and displacement. It is essential that these two channels share the same time steps and window size. The window size, which defines the length of the input sequence, is an adjustable parameter determined by the user or engineer.

Next, the **1D-CNN Block** is employed to extract features from the input data while simultaneously reducing computational costs. The 1D-CNN operates by scanning over the input data using kernels, effectively capturing relevant features. The computational cost of the model increases exponentially with the window size (i.e. the number of time steps inputted into the model). As a result, adding more parameters not only heightens computational demands but also increases model complexity, potentially hindering the model's convergence to its optimal state during training. Moreover, the convolutional operation in the 1D-CNN serves a dual purpose: extracting feature maps and reducing noise. Given the high-speed train operating environment and the inherent noise from sensors, the input data can be significantly noisy and the 1D-CNN is particularly effective in mitigating this issue.

Reducing noise in the data is crucial for significantly enhancing the quality of the extracted features. The 1D-CNN operates similarly to the 2D-CNNs commonly used in computer vision and other applications involving 2D data. The 1D-CNN utilises kernels that slide over the data to extract features. Let the input to the 1D-CNN be:

$$x = [x_1, x_2, \dots, x_L], \quad \text{where } x_i \in R^C \quad (2)$$

where m denotes the length of the input sequence, and C denotes the number of input channels. Let k be the kernel size, then the convolutional kernel is $w \in R^{k \times C}$, and the convolution output feature map at position i is:

$$y_i = \sum_{j=0}^{k-1} \sum_{c=1}^C w_{j,c} \cdot x_{(i+j),c} + b \quad (3)$$

where $w_{j,c}$ is the weight at position j for channel c , and b denotes the bias term. Therefore, the extracted feature maps can be represented by the full output of the convolution results $y = [y_1, y_2, \dots, y_T]$.

The learning process for a 1D-CNN is analogous to that of a standard neural network, where we can employ the gradient descent algorithm and backpropagation through the CNN layers. Specifically, the process utilises the chain rule to derive the partial derivative of the weights w_k with respect to the loss function $\mathcal{L}$, which quantifies the sensitivity of the model's predictions to errors.

The adjustment of the neuron weights during training can be expressed as:

$$\frac{\partial \mathcal{L}}{\partial w_k} = \sum_{j=0}^m \frac{\partial \mathcal{L}}{\partial y_j} \frac{\partial y_j}{\partial w_k}$$

Further, we can simplify the expression as:

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial y} * \overleftarrow{x}_x \quad (5)$$

whereas $\overleftarrow{x}_x$ is the flipped matrix of x , and $*$ denotes the convolve operation, and update the weights w_k with the learning rate α , and change $\frac{\partial \mathcal{L}}{\partial w_k}$.

$$w'_k \leftarrow w_k - \alpha \frac{\partial \mathcal{L}}{\partial w_k} \quad (6)$$

Third, the RNN block is designed to handle sequential or time-series data, where $\overrightarrow{a}^{<0>} \dots \overrightarrow{a}^{<T_x+1>}$ denotes activation functions over the time steps from T_0 to T_x in Figure 3. In this work, we specifically adopt the Gated Recurrent Unit (GRU) network as the backbone of our model. While Long Short-Term Memory (LSTM) networks generally outperform most RNN variants in modelling sequential data, we chose GRU due to its ability to effectively capture hidden sequential dependencies while mitigating the vanishing gradient problem. Moreover, GRUs offer these advantages with lower complexity and reduced computational costs compared to LSTMs, making them more suitable for our specific problem.

As illustrated in the Figure 4, each GRU neuron consists of several key components: the '+' (addition) operation, the σ (sigmoid) function, the 'tanh' activation function and the $\odot$ (Hadamard product). Additionally, each neuron contains an update gate and a reset gate to regulate the flow of information through the neuron. These gates control which information is retained or discarded as it passes through the neuron, functioning similarly to the gates in an LSTM but with a simpler structure, as GRUs have fewer gates.

The update z_t at time step t is defined as follows:

$$z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1}) \quad (7)$$

When the input x_t is fed into the network unit, it is multiplied by its corresponding weight $W^{(z)}$. Similarly, the hidden state from the previous time step, $h_{(t-1)}$, which carries information from previous units, is multiplied by its associated weight $U^{(z)}$. The sum of these two products is then passed through a sigmoid activation function, which squashes the result to a value between 0 and 1. The update gate plays a crucial role in helping the model determine how much of the past information (from previous time steps) should be retained

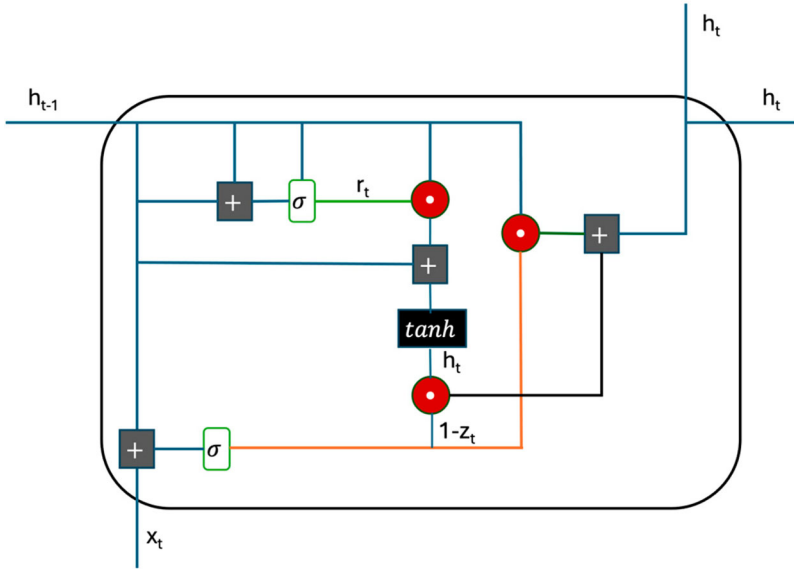


Figure 4. Gated recurrent unit.

and passed on to future time steps. This capability is particularly powerful as it allows the model to potentially carry forward all the relevant information from the past, effectively mitigating the risk of vanishing gradient problems. Essentially, this gate is used from the model to decide how much of the past information to forget.

$$r_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1}) \quad (8)$$

This formula mirrors the one used for the update gate, with the primary difference lying in the weights and the specific purpose of the gate. As before, the hidden state h_{t-1} and the input x_t are plugged in, multiplied by their respective weights, summed and then passed through the sigmoid function. To understand the influence of the gates on the final output, we begin by examining the role of the reset gate. We introduce a new memory content, which leverages the reset gate to selectively store relevant information from the past. This new memory content is calculated as follows:

$$h'_t = \tanh(Wx_t + r_t \odot Uh_{t-1}) \quad (9)$$

The process begins by multiplying the input x_t by a weight matrix $\mathbf{W}$, and similarly, multiplying the hidden state h_{t-1} by a weight matrix $\mathbf{U}$. Next, the reset gate r_t plays a crucial role by determining which information from previous time steps should be discarded. This is achieved through the Hadamard (element-wise) product between r_t and Uh_{t-1} . To accurately gauge the overall satisfaction, the model must focus on the final portion of the review. In the context of contact force prediction, the reset gate r_t in the GRU architecture modulates the contribution of past hidden states when generating new memory content. Specifically, as the model processes time-series input toward the most recent time steps, it can learn to reduce the influence of earlier time steps by decreasing the value of r_t , thereby prioritising recent, more informative segments of acceleration and uplift data. This gating mechanism enables the network to selectively filter outdated or less relevant information, enhancing its ability to capture the dynamic behaviour of the pantograph-catenary

interaction during high-speed operation. The training process for the GRU block is similar to that of a standard neural network, so we will not revisit those details here.

Finally, the network includes a fully connected layer, which processes the features and hidden information captured by the GRU to generate the final prediction. Given that the output (force) is a numerical value, the mean square error (MSE) is naturally chosen as the loss function, defined as:

$$L = \frac{1}{n} \sum_{i=1}^n (f - \hat{f})^2 \quad (10)$$

where L is the loss, n is the size of the sample data f is the true force and $\hat{f}$ is the predicted value.

The training algorithm presented as follows outlines the process for training a CNN-GRU model for a pantograph-catenary system (PCS).

Algorithm 1: CNN-GRU Model for PCS

Training:

```

// Run the model
while Trigger == True do
    Input: data, parameters
    Slice Displacement and Acceleration;
    Output: loss, gradients
    //loss: Meas Square Error
    //gradients: gradients of input-to-hidden
    weights
    //Forward propagate through time for both
    directions:
    Model_Forward:
        Input: data, parameters
        Output: hidden_states, prediction
        for iterate all time slots do
            forward propagation from CNN Block,
            GRU Block to NN Block;
            Save the value of the prediction force
            values;
        store values for backward propagation;
        update predict results;
    //Backpropagate through time for both
    directions:
    Model_Backward:
        Input: gradients, caches
        Output: gradients
        while iterate all time slots do
            compute gradients of parameters in NN
            Block;
            compute gradients of parameters in
            GRU Block;
            compute gradients of parameters in
            CNN Block;
        Store the gradients in a dictionary;
    Update parameters with the gradients and
    Adam;

```

The algorithm begins by running the model in a loop controlled by a trigger condition. During each iteration, the model slices the input data, which includes displacement and acceleration, and then performs forward propagation. The forward pass involves processing data through the CNN block, GRU block and fully connected neural network (NN) block to predict force values. The results, including hidden states and predictions, are stored for later use in the backward pass. After completing the forward pass, the algorithm proceeds with backward propagation to update the model parameters. This involves calculating gradients for each time step, starting from the fully connected layer and moving backward through the GRU and CNN blocks [49]. These gradients are accumulated and stored in a dictionary. Finally, the model parameters are updated using the Adam optimiser, which leverages the computed gradients to minimise the loss function, specifically the mean square error (MSE). This iterative process continues until the training is complete, optimising the model to accurately predict force based on the input data.

3. Numerical model of pantograph-catenary

The catenary is a tensioned cable structure that exhibits strong nonlinearities under the influence of the high-speed pantograph. A typical catenary system primarily comprises a messenger wire, a contact wire, stitch wires, droppers and steady arms, as shown in Figure 5. The contact wire, under significant tension, directly interacts with the pantograph to transmit electrical current to the train. The messenger wire and droppers suspend the contact wire, ensuring it stays in the correct position. Steady arms are placed at the end of each span to achieve a specific stagger value for the contact wire, while stitch wires are used in some catenary systems to compensate for elasticity irregularities. At speeds exceeding 350 km/h, key factors to consider include geometric nonlinearity, accurate representation of the contact wire geometry and the inclusion of the 3D steady arm. Previous models designed for conventional speeds [50] often neglected geometric nonlinearity since the vibrations of the contact wire were relatively small. However, at higher speeds, ignoring this can result in significant numerical errors, as the vibrations become more pronounced. In this paper, the catenary is modelled using the absolute nodal coordinate formulation (ANCF), a variant of the traditional finite element method (FEM) that effectively handles

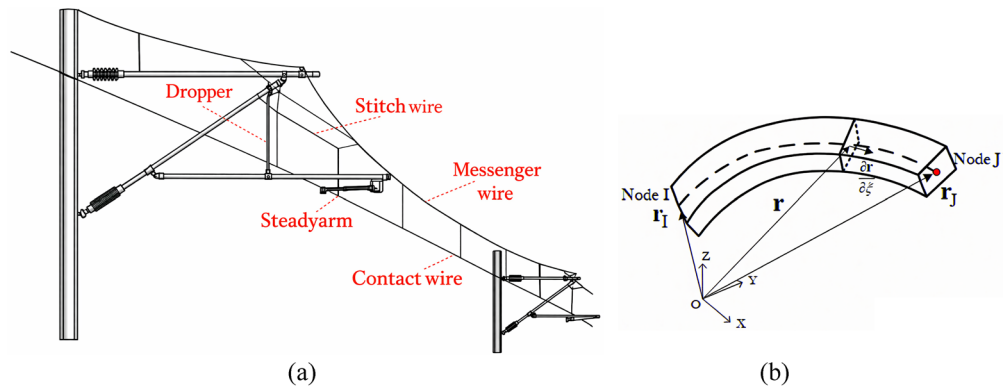


Figure 5. Catenary model (a) and ANCF beam element with two nodes I and J (b).

nonlinearity in large deformations [51]. Figure 5 illustrates an ANCF beam element used to model the contact wire, messenger wire and stitch wire. The dropper and steady arm are modelled using a similar element, but without bending degrees of freedom. Claws and clamps at connection points are assumed to be lumped masses attached to the wire. For an ANCF beam element with nodes I and J, the nodal degree of freedom (DOF) vector includes both displacements and gradients.

$$\mathbf{e} = \left[x_i \quad y_i \quad z_i \quad \frac{\partial x_i}{\partial \chi} \quad \frac{\partial y_i}{\partial \chi} \quad \frac{\partial z_i}{\partial \chi} \quad x_j \quad y_j \quad z_j \quad \frac{\partial x_j}{\partial \chi} \quad \frac{\partial y_j}{\partial \chi} \quad \frac{\partial z_j}{\partial \chi} \right]^T \quad (11)$$

where χ is the local coordinate along the beam ranging from 0 to the undeformed element length l_0 . The global position vector $\mathbf{r}$ can be expressed by

$$\mathbf{r} = \mathbf{S}\mathbf{e} \quad (12)$$

where $\mathbf{S}$ is the shape function matrix. The strain energy is calculated by the summation of the contributions of axial and bending deformations, which is expressed as

$$U = \frac{1}{2} \int_0^{L_0} (EA\varepsilon_l^2 + EI\kappa^2) d\chi \quad (13)$$

in which E is the Elastic modulus, A is the cross-sectional area, I is the moment of inertia. The axial strain ε_l and the curvature k can be expressed by [52]

$$\varepsilon_l = \frac{1}{2} \left(\frac{\partial \mathbf{r}}{\partial \chi}^T \frac{\partial \mathbf{r}}{\partial \chi} - 1 \right) = \frac{1}{2} \left(\mathbf{e}^T \frac{\partial \mathbf{S}}{\partial \chi}^T \frac{\partial \mathbf{S}}{\partial \chi} \mathbf{e} - 1 \right) \quad (14)$$

$$\kappa = \left| \frac{\partial^2 \mathbf{r}}{\partial \chi^2} \right| = \sqrt{\mathbf{e}^T \frac{\partial^2 \mathbf{S}}{\partial \chi^2}^T \frac{\partial^2 \mathbf{S}}{\partial \chi^2} \mathbf{e}} \quad (15)$$

The generalised elastic forces $\mathbf{Q}$ can be defined as

$$\mathbf{Q} = \frac{\partial U}{\partial \mathbf{e}} = \mathbf{K}_e \mathbf{e} \quad (16)$$

Thus, the secant matrix $\mathbf{K}_e$ with respect to the absolute coordinate can be obtained. In the shape-finding procedure, the tangent stiffness matrix is preferred to iteratively calculate the incremental nodal DOF vector $\Delta \mathbf{e}$ and the incremental unstrained length Δl_0 . The tangent stiffness matrices $\mathbf{K}_T$ and $\mathbf{K}_L$ can be obtained by the derivative of Eq. (7) with respect to $\mathbf{e}$ and l_0 as

$$\Delta \mathbf{F} = \frac{\partial \mathbf{Q}}{\partial \mathbf{e}} \Delta \mathbf{e} + \frac{\partial \mathbf{Q}}{\partial L_0} \Delta L_0 = \mathbf{K}_T \Delta \mathbf{e} + \mathbf{K}_L \Delta L_0 \quad (17)$$

Eq. (9) is the force balance equation for a single ANCF beam element. With an FE approach and a TCU shape-finding procedure, as illustrated in [53], the initial configuration of the catenary can be obtained. The TCU methodology, proposed by Kim and Lee [54], is employed to initialise the catenary. This involves treating the undeformed length as unknown parameters and creating tangent stiffness matrices, denoted as $\mathbf{K}_T$ and $\mathbf{K}_L$. However, a direct resolution of this issue is hindered by the non-square nature of $\begin{bmatrix} \mathbf{K}_T & \mathbf{K}_L \end{bmatrix}$.

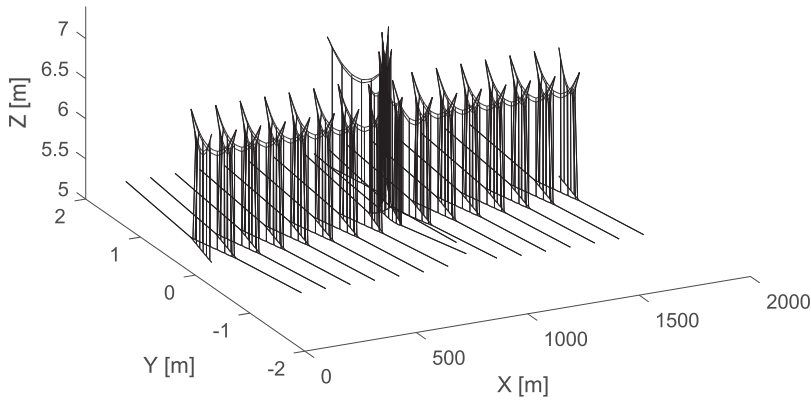


Figure 6. Geometry of the catenary modelled by ANCF beam elements.

The total number of unknowns surpasses the total number of equations, resulting in indeterminate solutions. To address this challenge, additional constraint conditions must be introduced to curb undesired movements based on design specifications. This process includes more additional constraints to transfer $[K_T \ K_L]$ into a square matrix, thereby ensuring the uniqueness of solutions. The global catenary geometry calculated by the present method is shown in Figure 6. To reproduce a realistic dataset, the catenary used in the simulation contains two tensile sections with one overlap section. After introducing a consistent mass matrix M_C^G and Rayleigh damping matrix C_C^G , the equation of motion for the catenary system can be written as

$$M_C^G \ddot{U}_C(t) + C_C^G \dot{U}_C(t) + K_C^G(t) U_C(t) = F_C^G(t) \quad (18)$$

in which K_C^G is the stiffness matrix of the catenary system, respectively. The vector $U_C(t)$ is the global DOF vector and $F_C^G(t)$ on the right-hand hand is the external force vector. The pantograph was modelled as a lumped-mass-spring-damper system with a simplified panhead geometry that interacts with the catenary through a penalty-based contact model. The panhead mass, spring stiffness and damping coefficients were calibrated based on typical characteristics of operational pantographs used in China's high-speed rail [55]. The present model has been validated against the benchmark and field test data across various speed levels [12,56,57]. Figure 7 shows one example of the resulting dynamic responses (including the panhead acceleration, uplift and contact force) of pantograph-catenary simulated via the proposed model at 350 km/h. Even from the time history, a high dependence of the contact force on the panhead acceleration/uplift can be observed. In the next section, four training strategies are proposed to test the prediction performance, incorporating variations in structural parameters, speed levels and cut-off frequencies.

4. Strategy for training and testing

Building on the numerical model developed in the previous section, a database is created, encompassing various catenary systems, operating speeds and cut-off frequencies. To

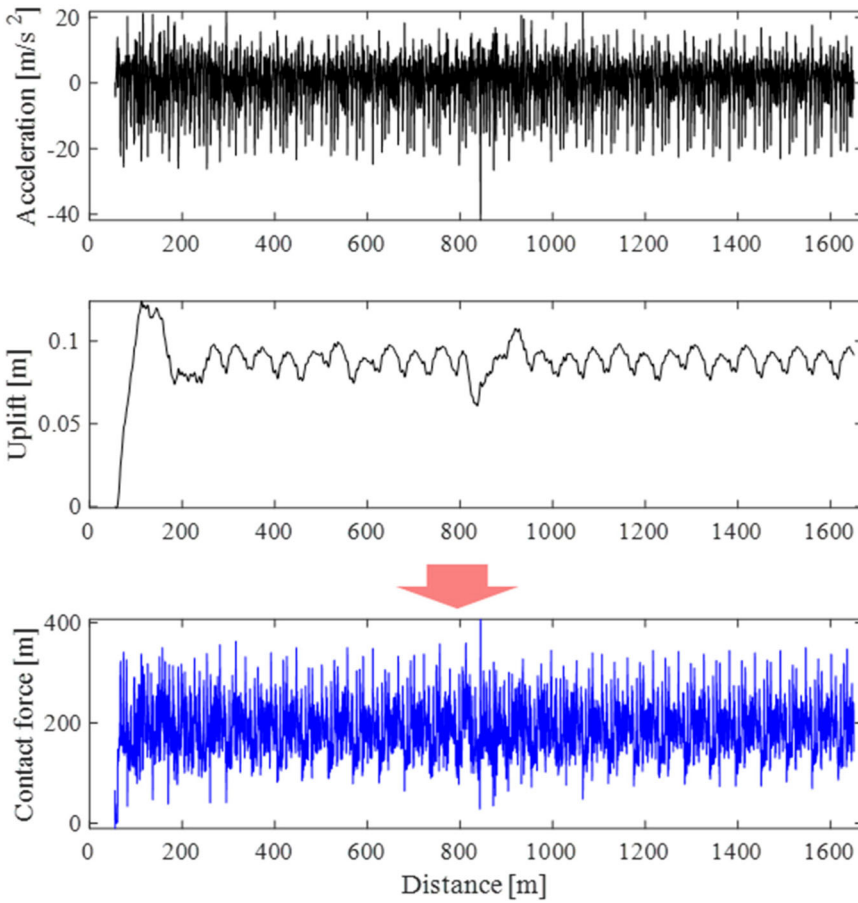
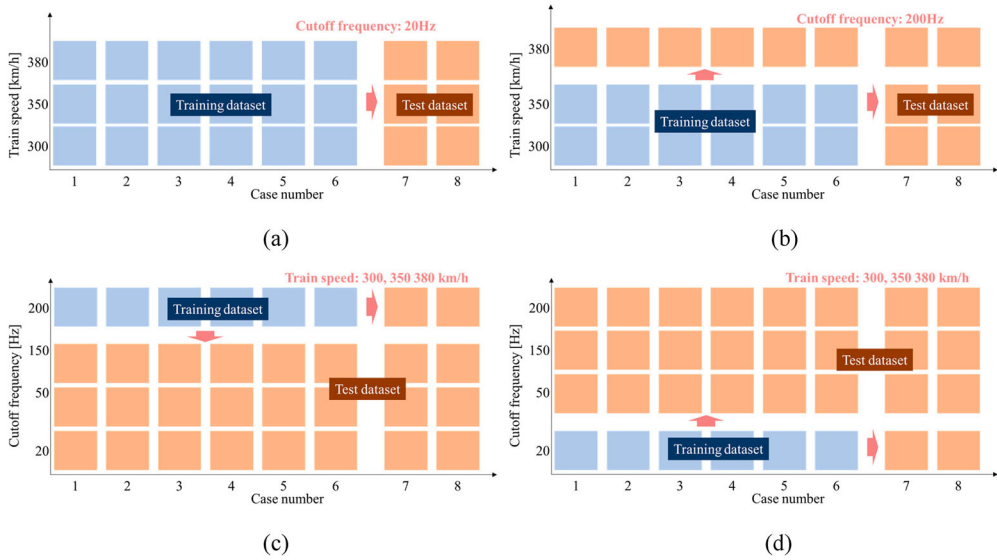


Figure 7. Dynamic response of pantograph-catenary. From top to bottom are panhead acceleration, uplift and contact force.

simulate various data acquisition scenarios and evaluate the model's generalisability, we applied low-pass filtering to the acceleration and uplift signals using cut-off frequencies of 20, 50, 150 and 200 Hz. These filtered signals serve as inputs to the deep learning model. The cut-off values reflect both industry standards (e.g. EN 50317 recommends 20 Hz) and practical considerations for capturing high-frequency dynamics. By varying the cut-off frequency, we construct datasets that differ in information content and noise levels, allowing us to assess how well the model generalises across different sensing conditions [5]. In this study, four cut-off frequency levels – 20, 50, 150 and 200 Hz – are considered. Additionally, three operating speeds – 300, 350 and 380 km/h – are used. The numerical simulations include eight different catenary configurations, each varying in span length, contact wire tension, messenger wire tension, number of droppers, stitch wire length and stitch wire tension, as detailed in Table 1. These eight catenary systems are broadly categorised into three types based on span length, messenger wire tension, contact wire tension, dropper count and stitch wire length, with two variations in stitch wire tension.

Table 1. Eight cases of catenary with different parameters.

No	SP (m)	TCW (N)	TMW (N)	ND	SWL (m)	TSW (N)
1	50	27000	23000	7	18	3500
2	45	27000	25000	6	14	2800
3	55	27000	23000	7	16	2800
4	55	28500	25000	8	14	3500
5	55	30000	21000	7	18	3500
6	50	28500	25000	7	14	3500
7	45	30000	21000	7	16	3500
8	50	30000	21000	7	18	3500

**Figure 8.** Training and testing strategies. (a–d) The training and testing dataset used in strategies 1–4.

Based on the aforementioned description, the database contains a total of 96 data groups (comprising 8 catenary systems, 3 speeds and 4 cut-off frequencies). Four training and testing strategies are implemented to evaluate the proposed deep learning framework, as illustrated in Figure 8(a–d). In the first strategy, as presented in Figure 8(a), only a cut-off frequency of 20 Hz is used. Data from six catenary cases are employed for training, while the remaining two cases serve as the test set. The second strategy, shown in Figure 8(b), increases the cut-off frequency to 200 Hz, allowing the framework to be tested across speed levels. Data from the first six catenary cases at speeds of 300 km/h and 350 km/h are used for training, with the rest reserved for testing. In the third strategy in Figure 8(c), the prediction performance across cut-off frequencies, from high to low, is examined. Data from the first six catenary cases at a cut-off frequency of 200 Hz are used for training, while data from the remaining frequencies and three additional catenary cases are used for testing. Finally, in the fourth strategy, depicted in Figure 8(d), the prediction performance is assessed as the cut-off frequency is increased from low to high. Data from the first six catenary cases at a 20 Hz cut-off frequency are used for training, with the remaining cut-off frequencies and three additional cases used to test the neural network. Although Strategies Figure 8(a and d) both train the models at 20 Hz, they use different partitions of

the 20 Hz dataset: Strategy Figure 8(a) reserves a portion of the 20 Hz samples for same-frequency testing, whereas Strategy Figure 8(d) uses the full 20 Hz dataset for training to enable cross-frequency evaluation on higher-frequency inputs.

The four training strategies were designed to evaluate the model's robustness under variations in catenary structure, train speed and signal cut-off frequency. Strategy (a) establishes the baseline performance at 20 Hz, which aligns with current measurement standards. Strategies (c) and (d) test the model's ability to generalise across cut-off frequencies. Although Strategy (d) shares the same training data as Strategy (a), it is used to assess whether low-frequency training can support high-frequency prediction – an important scenario when sensor upgrades or regional differences introduce bandwidth mismatches. Strategy (b) combines cut-off and speed variation to test combined generalisation capacity, and our results indicate only marginal performance degradation, suggesting the model is resilient to moderate speed variation.

5. Analysis of prediction performance

To assess the feasibility of predicting the contact force using panhead acceleration/uplift, four different training strategies were employed to evaluate the performance of the proposed deep learning models. The mean square error (MSE) between the predicted and actual data was used as the primary metric for comparing model performance. To ensure consistency across different network architectures, all RNN-based models (RNN, LSTM, GRU) were constructed with two recurrent layers, each containing 32 neurons, followed by a fully connected output layer with a single neuron to predict the contact force. For the proposed CNN-GRU model, a one-dimensional convolutional layer with 64 neurons was added prior to the GRU layers to perform feature extraction and noise reduction, while the GRU component retains the same structure as the standalone GRU model. The resulting model sizes are 3,233 parameters for the RNN, 9,825 for the GRU, 12,833 for the LSTM and 15,969 for the CNN-GRU model. These compact architectures (< 0.1 MB) ensure a fair comparison across methods, and the additional convolutional layer in the CNN-GRU enhances local feature extraction and noise mitigation.

The exact dataset sizes may vary slightly depending on the specific training strategy. In general, the training set comprises approximately 124,200 samples, while the test set includes around 41,400 samples. After conducting several experiments on the simulation dataset, we found that setting the input time window size to 16 as the time step size provided a good balance between model performance and computational efficiency. This window length was sufficient to capture the relevant temporal dynamics in the pantograph-catenary interaction, while avoiding excessive sequence length that would increase the number of parameters in the recurrent layers and lead to higher computational cost. Figure 9(a and b) shows the prediction performance at all train speeds for same-frequency evaluation, whereas Figure 9(c and d) presents only the cross-frequency results at frequencies different from the respective training frequency. The same-frequency cases (20 Hz for Strategy 1 and 200 Hz for Strategy 2) are not repeated in Figure 9(c and d) because they are already shown in Figure 9(a and b). Using one training strategy (Figure 8(a)) as an example, the learning curves of four methods are shown in Figure 10. Among these, the CNN-GRU model clearly outperforms the others. While the loss curve of the proposed model exhibits

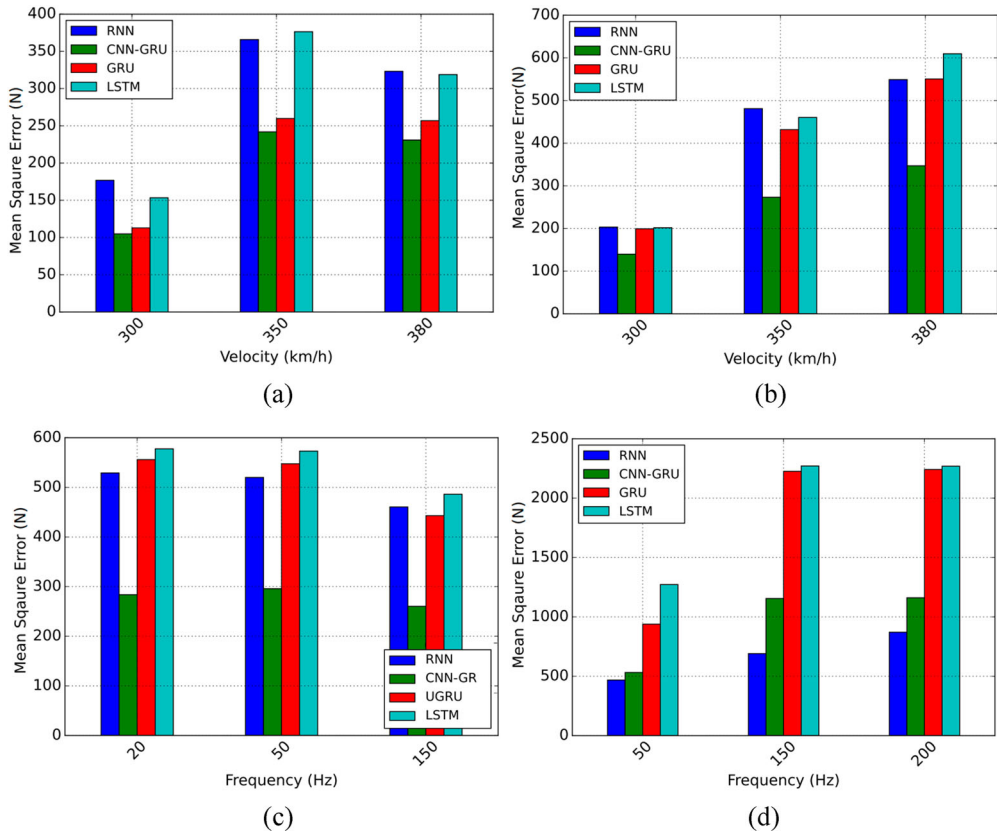


Figure 9. Comparison of prediction accuracy using (a) training strategy 1; (b) training strategy 2; (c) training strategy 3; (d) training strategy 4.

some oscillation due to the randomness in the initialisation of parameters and the training dataset, it consistently demonstrates lower loss (error) compared to the other models.

Figure 9 summarises the prediction performance of the four models – RNN, GRU, CNN-GRU and LSTM – on the test data across the four training strategies, whereas Figure 10 illustrates the validation loss curves during training under Strategy 2, highlighting the convergence behaviour of each model. Among these, the proposed model (CNN-GRU) outperforms the rest of the model, achieving the lowest validation loss of 180.24 at epoch 18, indicating its superior ability to capture temporal dependencies and complex patterns in the data. The GRU model also performed well, with a validation loss of 265.11 at the same epoch, showcasing its efficiency in handling sequential data. In contrast, the RNN model exhibited the highest validation loss of 315.56 at epoch 18, while the LSTM model achieved a lower loss, suggesting they were less effective in learning the underlying patterns of the dataset, though still higher than GRU and CNN-GRU. These results highlight the advantage of combining convolutional layers with GRUs in the CNN-GRU model, as it effectively balances feature extraction and sequential learning, leading to more accurate predictions. Regarding the prediction accuracy across different strategies, a detailed analysis is provided below:

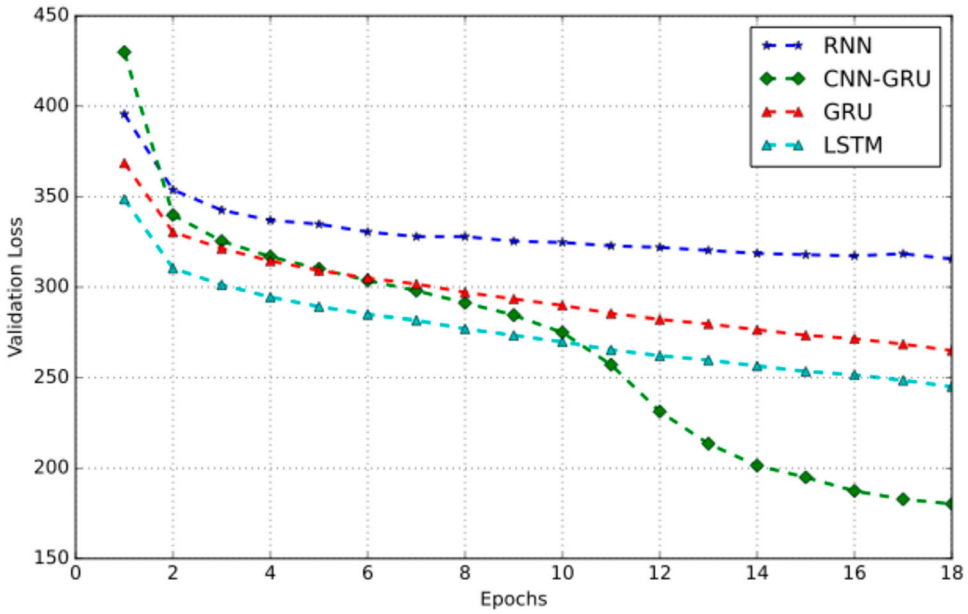


Figure 10. Validation loss curve of the four methods.

5.1. Strategy 1

In Strategy 1, the goal is to assess the feasibility of using panhead acceleration and uplift within the 0–20 Hz frequency range to predict contact force at the same cut-off frequency, as outlined in Section IV. The 0–20 Hz range is critical for fully describing pantograph-catenary interaction performance, according to established assessment specifications [5]. In this scenario, four networks were trained using data from Catenary Cases 1–6 at speeds of 300, 350 and 380 km/h. The remaining two cases (Cases 7 and 8) were used as test data. The MSE results for the four methods are shown in Figure 9(a). While all methods perform best at 300 km/h, their accuracy declines as the train speed increases as train speed increases. This is likely because the models were trained on lower-speed data, yet the results still demonstrate good generalisation capability, as the models remain effective under higher-speed conditions not seen during training. Among the methods, GRU and CNN-GRU consistently outperformed the others. To visualise the prediction performance, two examples – ‘Case 7 + 300 km/h’ and ‘Case 8 + 380 km/h’ – are shown in Figure 11. The predicted contact force closely matches the actual force, with CNN-GRU yielding the most accurate time history, especially at peak values.

5.2. Strategy 2

In Strategy 2, the cut-off frequency is raised to 200 Hz, and the prediction performance across different speeds is evaluated. Data from the first six catenary cases at 300 and 350 km/h were used for training, while the remaining dataset was used for testing. The MSE results are presented in Figure 9(b), and the findings are consistent with Strategy 1, though the overall error increased with the higher cut-off frequency. Two examples –

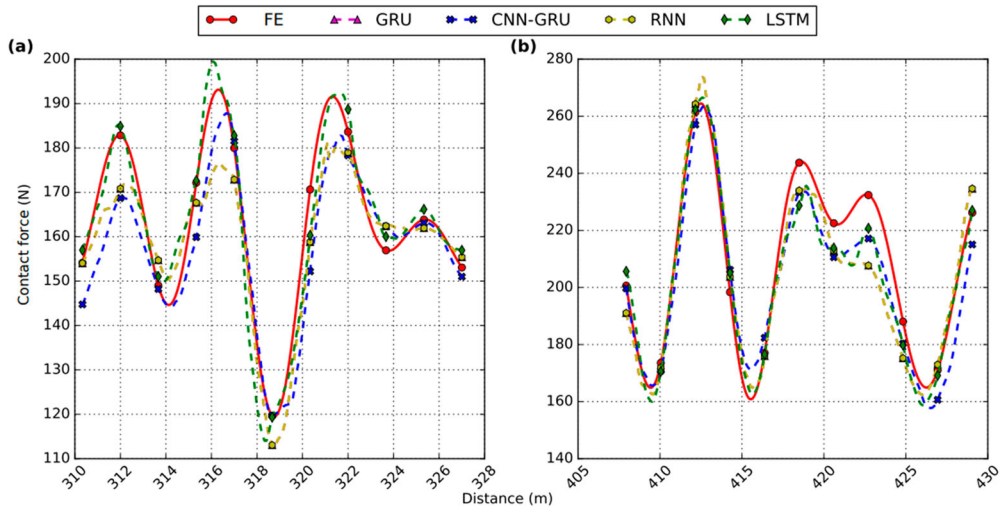


Figure 11. Two examples of resulting prediction contact force using training strategy 1: (a) 300 km/h + case 7; (b) 380 km/h + case 8.

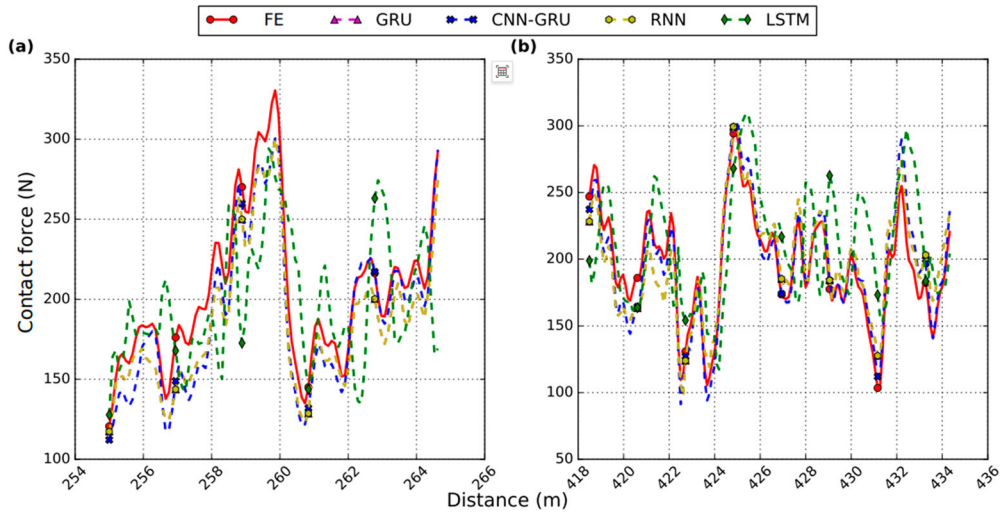


Figure 12. Two examples of resulting prediction contact force using training strategy 2: (a) 350 km/h + case 7 + 200 Hz; (b) 380 km/h + case 8 + 200 Hz.

‘Case 7 + 350 km/h + 200 Hz’ and ‘Case 8 + 380 km/h + 200 Hz’ – are shown in Figure 12. Although the neural network still captures the overall waveform of the contact force, larger errors are noticeable at some peak values compared to those in Figure 12. Despite the higher frequency introducing greater uncertainty, the model’s prediction remains acceptable, as it effectively captures high-frequency characteristics of the interaction.

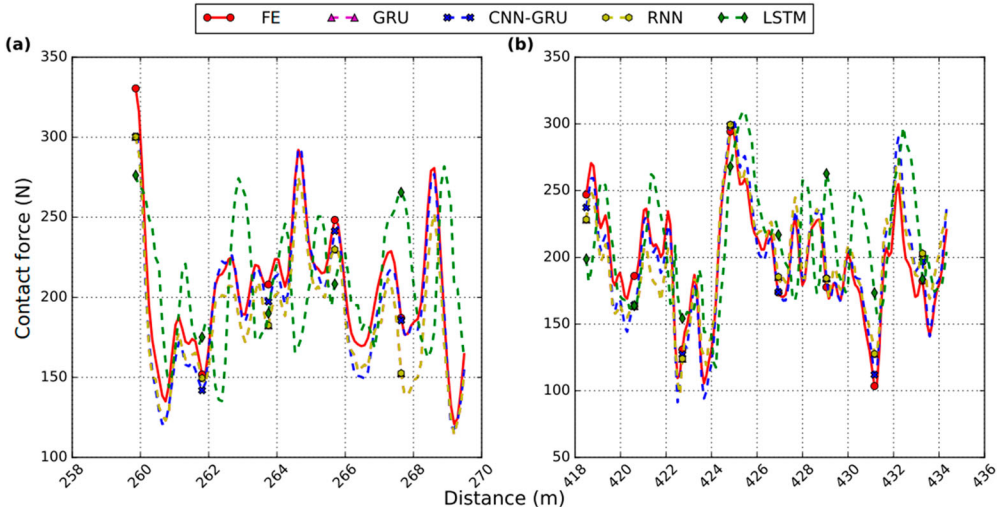


Figure 13. Two examples of resulting prediction contact force using training strategy 3: (a) 350 km/h + case 7 + 50 Hz; (b) 380 km/h + case 8 + 150 Hz.

5.3. Strategy 3

Strategy 3 tests the prediction performance across cut-off frequencies from high to low. Data from the first six catenary cases with a 200 Hz cut-off frequency were used for training, and data with lower cut-off frequencies served as the test set. The MSE results, shown in Figure 9(c), indicate that overall error levels are comparable to those in Strategy 2, with CNN-GRU delivering the best performance. Two examples – ‘Case 7 + 350 km/h + 50 Hz’ and ‘Case 8 + 380 km/h + 150 Hz’ – are shown in Figure 13. The neural network generally captures the fluctuations in contact force, but a larger discrepancy is evident in Figure 13, indicating that prediction accuracy deteriorates at higher speeds and higher cut-off frequencies.

5.4. Strategy 4

In Strategy 4, the prediction performance is evaluated across cut-off frequencies from low to high. Data from the first six catenary cases with a 20 Hz cut-off frequency were used for training, and data with higher cut-off frequencies were used for testing. The MSE results in Figure 9(d) reveal significantly higher error rates compared to the previous three strategies. Except for RNN and CNN-GRU, the errors of the other models exceed 1000 N, indicating poor performance. Two examples – ‘Case 7 + 350 km/h + 150 Hz’ and ‘Case 8 + 380 km/h + 200 Hz’ – are shown in Figure 14. A noticeable discrepancy between the predicted and actual results can be observed, particularly in Figure 14(b), highlighting the degradation in performance with increasing frequency and speed. The observed performance degradation when applying models trained on high-frequency data (e.g. 200 Hz) to lower-frequency test data highlights the sensitivity of deep learning models to the spectral characteristics of the input signals. Different cut-off frequencies capture varying levels of dynamic detail in the pantograph-catenary interaction. In particular, models trained

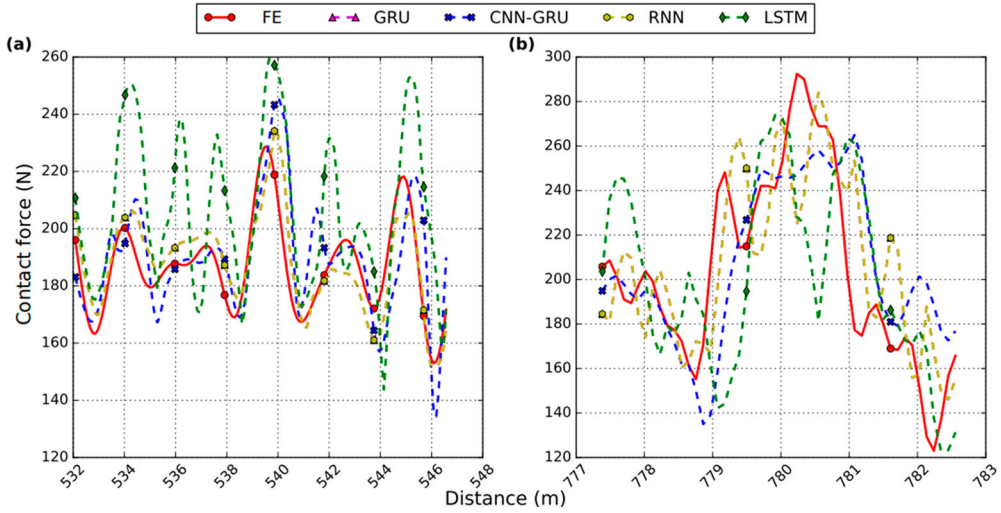


Figure 14. Two examples of resulting prediction contact force using training strategy 4: (a) 350 km/h + case 7 + 150 Hz; (b) 380 km/h + case 8 + 200 Hz.

on lower-frequency data may struggle to generalise to higher-frequency data, where rapid fluctuations and transient responses become more prominent. This limitation reflects a key challenge when applying such models to real-world scenarios, where measurement conditions and signal bandwidths may vary significantly.

6. Conclusions

This work explores the possibility of using a deep learning framework to predict the pantograph-catenary contact force based on the panhead acceleration and uplift. Three common neural networks (RNN, LSTM and GRU) and one developed network called CNN-GRU are used to check their performance in predicting the contact force. The main conclusions can be drawn as follows:

- (1) This preliminary study demonstrates the potential of using deep learning to predict contact force based on panhead acceleration, with promising results from all tested models.
- (2) The trained models can provide reliable predictions across different speed ranges, though accuracy tends to decline as speed increases.
- (3) Prediction performance degrades when applying models trained at one cut-off frequency to datasets with different frequencies, particularly at higher frequencies.
- (4) While RNN, LSTM and GRU show acceptable prediction accuracy, the CNN-GRU model outperforms the others, particularly in terms of convergence speed and overall accuracy, making it the most effective approach for this problem.

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No potential conflict of interest was reported by the author(s).

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